Vorsu Rajeswari

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
Narayana junior college	Class XII (BIEAP)	2019	9.87%
Narayana school	Class X (BSEAP)	2017	10.00%

SUMMARY

M.Tech Chemical Engineering student with a B.Pharm background and knowledge of Python, SQL, data analysis, statistical analysis, and machine learning. Strong analytical and problem-solving abilities, with an interest in applying data and technology across diverse domains.

PROJECTS

Optimizing Sensory Attributes and Stability of Cosmetic Emulsions through Microstructure and Rheological Engineering

- Formulated cosmetic O/W emulsions and assessed their behavior during storage.
- Characterized emulsion microstructure and creaming behavior using optical microscopy and image analysis.
- Investigated the influence of formulation composition, surfactant system, and processing conditions on microstructure and performance.
- Planning rheological characterization to evaluate flow behavior and its relationship with microstructure and stability.

Machine Learning-Based Prediction of Pharmaceutical Nanomedicine Performance

- Analyzed pharmaceutical nanomedicine data to identify relationships between formulation properties and performance.
- Performed data cleaning, exploratory data analysis, and visualization using Python.
- Applied data preprocessing and normalization to prepare the dataset for machine learning.
- Developed machine learning models to predict pharmaceutical nanomedicine performance.
- Evaluated model performance using R^2 and Mean Squared Error (MSE).
- Analyzed model results to identify key factors influencing predicted performance.

Acetic Anhydride Production Simulation and Economic Evaluation

- Compared alternative production routes based on yield, energy demand, and economic feasibility, selecting methyl acetate carbonylation as the preferred route.
- Simulated the process in Aspen Plus using CSTR, reactive distillation, flash, and RadFrac models.
- Achieved 92% methanol conversion and 99.9% methyl acetate purity with an estimated 3.17-year payback period.

Herbal Sunscreen Lotion - Formulation and Evaluation

- Developed a herbal sunscreen lotion using natural ingredients and optimized the formulation for desired properties.
- Evaluated pH, viscosity, spreadability, washability, skin irritation, physical stability, and SPF.
- Obtained a stable formulation with suitable application properties and an SPF of 25.

CERTIFICATIONS AND LICENSES

Adani FutureX Program - Industrial Visit, Adani Cement, Chittapur

Observed end-to-end cement manufacturing operations, from limestone mining, raw material handling, crushing, raw meal preparation, kiln processing, and clinker formation to cement grinding, packing, and dispatch.

Goldfish Pharma Private Limited

Completed 150 hours of industrial training, gaining exposure to GMP practices, pharmaceutical manufacturing operations, and industry work environment.

PUBLICATIONS

Preparation and Evaluation of Herbal Sunscreen Lotion

International Journal of Pharmacognosy and Chemistry (IJPC)

Developed a multi-component herbal sunscreen lotion and evaluated its physicochemical and photoprotective properties, including pH, viscosity, spreadability, homogeneity, washability, irritancy, and SPF.

SKILLS

Programming & Data Analysis: Python, SQL, Pandas, NumPy, Scikit-learn, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA).

Statistical Analysis: Descriptive Statistics, Hypothesis Testing, Correlation Analysis, ANOVA, Principal Component Analysis (PCA), Statistical Modelling.

Machine Learning: Regression, Classification, Clustering, Predictive Modelling, Model Evaluation.

Data Visualization: Excel, Matplotlib, Origin, Data Visualization.

Pharmaceutical & Formulation: Pharmaceutical Formulation, Emulsion Stability, Droplet Size Analysis, Microstructure Analysis, Microscopy, ImageJ.

Process Engineering: Aspen Plus, ANSYS Fluent, ANSYS Meshing, Heat Integration, Sensitivity Analysis.